

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	01 October 2026
Team ID	6ab7f80ad083b0e2ee72b17e
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Diagnostic Learning Readiness & Gap Detection	Short diagnostic questions to estimate readiness Prerequisite gap detection before starting a topic Learner readiness summary and starting-level recommendation
FR-2	AI-Assisted Study Plan Generation	Weekly study schedule from deadlines and subject priority Daily study blocks based on available time Plan rebalancing when a task is delayed
FR-3	Multi-Format Learning Presentation	Text plus visual representation of a topic Dense notes converted to structured visual outlines Examples, diagrams and notes; learner chooses format
FR-4	Learning Progress & Goal Monitoring	Topic completion and pending-work tracking Learner-defined goals, milestones and checkpoints Simple progress dashboard and summaries
FR-5	Context-Aware Academic Support	Detect the subject from the active learning context Session continuity for topic-linked follow-up answers Subject-specific guidance and tailored examples

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple, chat-based interface that students can use easily, with clear visual and structured outputs and minimal learning effort.
NFR-2	Security	Student data (progress, goals, study plans) is protected with secure authentication and protected API keys; only relevant data is shared with the Gemini API.
NFR-3	Reliability	Consistent and accurate responses across text, visual and structured formats, with graceful error handling if the AI service fails.
NFR-4	Performance	Responses to queries, diagnostics and study plans are generated within a few seconds under normal load.
NFR-5	Availability	The assistant is accessible on demand so students can study at any time, relying on a cloud-hosted service with high uptime.
NFR-6	Scalability	The system can support a growing number of students and subjects concurrently by using scalable cloud infrastructure and Gemini API capacity.